

Field and Galois Theory

1.

- (a) Give an example of a tower of field extensions $F \supset L \supset K$ such that L/K is Galois and F/L is Galois, but F/K is not Galois.
- (b) Give an example of a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Prove that the element you give is indeed a primitive element. Find the minimal polynomial of the element you find.

1. Solution

- (a) Consider $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{2})$, and $F = \mathbb{Q}(\sqrt[4]{2})$. Then $F \supset L \supset K$, and L/K is Galois since it's the splitting field of the separable polynomial $x^2 - 2 \in \mathbb{Q}[x]$, and F/L is Galois since it's the splitting field of the separable polynomial $x^2 - \sqrt{2} \in \mathbb{Q}(\sqrt{2})[x]$, but F/K is not Galois since it is not a splitting field (it doesn't contain the roots $\pm i\sqrt[4]{2}$ of the minimal polynomial of $\sqrt[4]{2}$, $m(x) = x^4 - 2 = (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - i\sqrt[4]{2})(x + i\sqrt[4]{2})$).
- (b) $\sqrt{2} + \sqrt{3}$ is a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Its minimal polynomial is

$$m(x) = (x - (\sqrt{2} + \sqrt{3}))(x - (\sqrt{2} - \sqrt{3}))(x + (\sqrt{2} + \sqrt{3}))(x + (\sqrt{2} - \sqrt{3})) = x^4 - 10x^2 + 1.$$

Therefore, $\mathbb{Q}(\sqrt{2} + \sqrt{3})/\mathbb{Q}$ is a degree 4 extension, and $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$. But $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \leq 2$, so $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] \leq 4$. Thus, we must have equality, so $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

2. Let ζ be a primitive k th root of unity.

- (a) Prove that $K = \mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$.
- (b) Compute $\text{Gal}(K/\mathbb{Q})$.

2. Solution

- (a) Let $\Phi_k(x)$ be the k th cyclotomic polynomial, which is defined iteratively by

$$\Phi_k(x) = \frac{x^k - 1}{\prod_{d|k, d < k} \Phi_d(x)}.$$

Then $\Phi_k(x)$ is the minimal polynomial of ζ since it is irreducible over $\mathbb{Q}$. The roots of $\Phi_k(x)$ are ζ^i for each $1 \leq i \leq k$ with $\gcd(k, i) = 1$. Therefore, $K = \mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois since $\Phi_k(x)$ is separable and K contains all of the roots of $\Phi_k(x)$.

- (b) By part (a), each automorphism of $K/\mathbb{Q}$ is determined by $\zeta \mapsto \zeta^i$ for some $1 \leq i \leq k$ with $\gcd(k, i) = 1$. Therefore, $|\text{Gal}(K/\mathbb{Q})| = \varphi(k)$ and $\text{Gal}(K/\mathbb{Q}) = (\mathbb{Z}/k\mathbb{Z})^*$.

3. (January 2020 Problem 2)

- (a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.
- (b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F .
- (c) Compute the Galois group $\text{Gal}(F(b)/F)$.

3. Solution

- (a) The roots of $x^4 - 7$ are $\pm\sqrt[4]{7}$ and $\pm i\sqrt[4]{7}$. If $a = \pm\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $i\sqrt[4]{7}$ so it is not a splitting field, and thus not Galois. If $a = \pm i\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $\sqrt[4]{7}$, so again is not a splitting field so not Galois over $\mathbb{Q}$.
- (b) Since $x^4 - 7 \equiv x^4 - 2 \pmod{5}$, we get that the roots of this polynomial over F are $\pm b$ and $\pm ib$, where $i = \sqrt{-1}$. Again, since $\text{char}(F) = 5$, $i = \sqrt{4} = 2$, so the roots are $\pm b$ and $\pm 2b$, or, more simply, $b, 2b, 3b$, and $4b$. Since each of these is in $F(b)$, it is the splitting field of $x^4 - 7$, and since this polynomial is separable, $F(b)$ is Galois over F .
- (c) An automorphism $\sigma \in \text{Gal}(F(b)/F)$ is determined by which root it sends b to. Thus, the elements in $\text{Gal}(F(b)/F)$ are $\sigma_1 = \text{Id} : b \mapsto b$, $\sigma_2 : b \mapsto 2b$, $\sigma_3 : b \mapsto 3b$, and $\sigma_4 : b \mapsto 4b$. Observe that $\sigma_2^2 = \sigma_3^2 = \sigma_4$ and $\sigma_4^2 = \text{Id}$. Therefore, $\text{Gal}(F(b)/F) \cong (\mathbb{Z}/5\mathbb{Z})^*$ where the isomorphism is given by $\sigma_i \mapsto i$.

4. (January 2021 Problem 5) Let $\mathbb{F}_p$ denote the field with p elements.

- (a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?
- (b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a .
- (c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

4. Solution

- (a) Observe that $x^4 + 1 = \Phi_8(x)$, so the roots of this polynomial are primitive 8th roots of unity. There are no primitive 8th roots of unity in $\mathbb{F}_3$, but in $\mathbb{F}_9$ every element is a root of $x^9 - x$, so every nonzero element is a root of $x^8 - 1$, and so there are primitive 8th roots of unity. In particular, if $i^2 = -1$ in $\mathbb{F}_9$, then $\pm 1 \pm i$ are primitive 8th roots of unity. Pairing these up, we can find that $x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1)$.
- (b) Consider $a = 2$. Then $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$. Indeed, it has no linear factors since $\alpha^4 \in \{0, 1\}$ for all $\alpha \in \mathbb{F}_5$. To show it has no quadratic factors, we need to show that $x^4 - 2$ does not split over $\mathbb{F}_{25}$. For every $x \in \mathbb{F}_{25}$, we have that $x^5 \in \mathbb{F}_5$, so if $x \in \mathbb{F}_{25} \setminus \mathbb{F}_5$ and $x^5 = n \in \mathbb{F}_5$, then $x^4 = n/x \notin \mathbb{F}_5$. Thus, $x^4 - 2$ has no roots in $\mathbb{F}_{25}$, and hence has no quadratic factors over $\mathbb{F}_5$.

- (c) Consider the map $\text{pow}_4 : \mathbb{F}_{103}^\times \rightarrow \mathbb{F}_{103}^\times$ given by $x \mapsto x^4$. This is a group homomorphism, whose kernel contains ± 1 , and has order dividing 4 since $x^4 - 1$ has at most 4 roots. However, $\#\ker \text{pow}_4$ cannot be 4, since $\#\mathbb{F}_{103}^\times = 102$ is not divisible by 4, so $\ker \text{pow}_4 = \{\pm 1\}$. In addition, this also shows that -1 is not a 4th power in $\mathbb{F}_{103}$, which we will use in a moment. Then we have two cases: either $a \in \text{im } \text{pow}_4$ or $a \notin \text{im } \text{pow}_4$.

In the first case, a has exactly two preimages under pow_4 , so $x^4 - a$ factors as two linear factors times an irreducible quadratic factor.

In the second case, we want to show that $x^4 - a$ factors into two irreducible quadratic factors. This is the same as saying $x^4 - a$ splits over $\mathbb{F}_{103^2}$. We begin by noting that $\mathbb{F}_{103^2}/\text{im } \text{pow}_4 \cong \mathbb{Z}/2\mathbb{Z}$, and a and -1 both represent the nontrivial element of the quotient, so $-a \in \text{im } \text{pow}_4$. Now, consider the analogous map $x \mapsto x^4$ on $\mathbb{F}_{103^2}^\times$. For every $b \in \mathbb{F}_{103}$, there is an element $c \in \mathbb{F}_{103^2}$ satisfying $c^2 = b$, so in particular we can find elements $i = \sqrt{-1}$ and $\sqrt{2}$. This has two consequences: first, the kernel of pow_4 is now order 4; second, we have that the element $(1+i)/\sqrt{2} \in \mathbb{F}_{103}$, and this element satisfies $((1+i)/\sqrt{2})^4 = -1$, so -1 is in the image of this map now. We said above that $-a = d^4$ for some $d \in \mathbb{F}_{103}$, so now we find that $a = (d(1+i)/\sqrt{2})^4$ in $\mathbb{F}_{103^2}$. Thus, every $a \in \mathbb{F}_{103}$ is a 4th power in $\mathbb{F}_{103^2}$, and since the kernel of pow_4 is order 4, every nonzero a has 4 distinct preimages. So, $x^4 - a$ splits in $\mathbb{F}_{103^2}$ as desired.

(In fact, since $103 \equiv 7 \pmod{8}$, $\sqrt{2} \in \mathbb{F}_{103}$, and so we can factor $x^4 - a$ as $(x^2 + \sqrt{2}d + d^2)(x^2 - \sqrt{2}d + d^2)$.)

Below are alternative, more explicit computational solutions for (a) and (b):

- (a) Observe that $x^4 + 1$ has no roots in $\mathbb{F}_3$, so it has no linear factors over $\mathbb{F}_3[x]$. Therefore, if it factors it must be as $(x^2 + ax + b)(x^2 + cx + d)$ for some $a, b, c, d \in \mathbb{F}_3$. Multiplying out, we see that we must have $a + c = 0$, $ac + b + d = 0$, and $bd = 1$. The last equation only has solutions $b = d = 1$ and $b = d = 2$. The former gives no solutions to the remaining equations, and the latter gives the solutions $a = 1, c = 2$ and $a = 2, c = 1$. Thus, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$ over $\mathbb{F}_3[x]$.
- (b) Consider $a = 2$. Then $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$. Indeed, it has no linear factors since $\alpha^4 \in \{0, 1\}$ for all $\alpha \in \mathbb{F}_5$. So, we just need to show that it has no quadratic factors $x^2 + bx + c$ with $b, c \in \mathbb{F}_5$. Observe that if $x^2 = -bx - c$, then $x^4 = (-b^3 + 2bc)x - b^2c + c^2$. So, we need to show that if $-b^3 + 2bc = 0$, then $-b^2c + c^2 \neq 2$. If $b = 0$, then it is clear that $c^2 \neq 2$ for any choice of c . If $b \neq 0$, then the only possible solutions for $-b^3 + 2c = 0$ are $b \in \{1, 4\}, c = 3$ and $b \in \{2, 3\}, c = 3$. In the first case, $-b^2c + c^2 = 0$, and in the second case, $-b^2c + c^2 = 3$. Therefore, $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$.

5. (January 2016 Problem 1) Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f .

- (a) Show that if $\text{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.

- (b) Is the converse statement true? That is, suppose $L = K(\alpha)$; must $\text{Gal}(L/K)$ be abelian?

5. Solution

- (a) Assume that $G = \text{Gal}(L/K)$ is abelian. We know that $K(\alpha) \subseteq L$, so the Galois correspondence implies that there is a subgroup $H \leq G$ such that $H = \text{Gal}(L/K(\alpha))$. Since G is abelian, H is a normal subgroup, so $K(\alpha)$ is a Galois extension of K . Since $f(x)$ is irreducible, it is the minimal polynomial of α over K , so f splits in $K(\alpha)$ since $K(\alpha)/K$ is Galois and contains a root of f so must contain all of the roots of f . Thus, $K(\alpha) = L$.
- (b) This is not true. Let L be any finite Galois extension of $\mathbb{Q}$ with non-abelian Galois group. Then the primitive element theorem gives that $L = \mathbb{Q}(\alpha)$ for some $\alpha \in L$.

6. (January 2017 Problem 4) This is a question about “biquadratic extensions” in two parts.

- (a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\text{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E .
- (b) Give an example of a field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$

6. Solution

- (a) Suppose $F = E(\sqrt{x}, \sqrt{y})$, where none of x, y , and xy are squares in E . This is a degree 4 extension, since it has basis $1, \sqrt{x}, \sqrt{y}, \sqrt{xy}$. It has four easy-to-think-of automorphisms: the identity, σ which flips the sign of $\sqrt{x}$ and $\sqrt{xy}$, τ which flips the sign of $\sqrt{y}$ and $\sqrt{xy}$, and $\sigma\tau$ which flips the sign of $\sqrt{x}$ and $\sqrt{y}$. Since the automorphism group can have size at most 4, we know we’ve found all the automorphisms, so this is in fact a Galois extension.

On the other hand, suppose F/E is a degree 4 Galois extension with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. Then, by the Fundamental Theorem of Galois Theory, from the subgroup structure of $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, we know that F has three intermediate extensions, which are degree 2 over E . Since E is not characteristic 2, the quadratic formula tells us that any degree 2 extension is of the form $E(\sqrt{d})$ for some nonsquare d . So, we can say two of the intermediate extensions are $E(\sqrt{x})$ and $E(\sqrt{y})$. Since these are distinct extensions of E , $\sqrt{x} \notin E(\sqrt{y})$ and visa versa, so $E(\sqrt{x}, \sqrt{y})$ is a degree 4 extension of E sitting inside F , and thus in fact we have $F = E(\sqrt{x}, \sqrt{y})$.

- (b) $E = \mathbb{F}_2$ works, because all the extensions of $\mathbb{F}_2$ have cyclic Galois group. If you can find and prove an infinite field has this property I’ll be pretty impressed.

7. Suppose that $K/\mathbb{Q}$ is a degree-4 Galois extension with Galois group $\mathbb{Z}/4\mathbb{Z}$. Prove that $\sqrt{-1} \notin K$.

7. Solution Suppose that K is a degree 4 Galois extension, and that $\sqrt{-1} \in K$. Then $\mathbb{Q}(\sqrt{-1}) \subset K$ is a subfield. But also, if we embed K in $\mathbb{C}$, we can take the subfield of K fixed by complex conjugation. This subfield must be a degree-2 subextension of K , but it cannot be $\mathbb{Q}(\sqrt{-1})$, so K has two different degree-2 subextensions. This implies that $\text{Gal}(K/\mathbb{Q})$ has two different subgroups of order 2. But $\mathbb{Z}/4\mathbb{Z}$ has only one such subgroup, so $\text{Gal}(K/\mathbb{Q}) \neq \mathbb{Z}/4\mathbb{Z}$.

8. (August 2018 Problem 4)

- (a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.
- (b) Let $K = \mathbb{F}_q(t)$, and let $L = \mathbb{F}_q(t^{1/p})$ where q is a power of p . Prove that there are no automorphisms of L fixing K .

8. Solution

- (a) Consider $K = \mathbb{Q}(\zeta_5)$ where ζ_5 is a primitive 5th root of unity. Then by problem 2 we have that $\text{Gal}(K/\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z}^\times = \mathbb{Z}/4\mathbb{Z}$.
- (b) Observe that $t^{1/p}$ satisfies $x^p - t \in K[x]$, and because K is characteristic p , this polynomial factors as $x^p - t = x^p - (t^{1/p})^p = (x - t^{1/p})^p$. Therefore, $t^{1/p}$ is its only root. Since L is the splitting field of $x^p - t$ over K , any automorphism of L over K is determined by how it permutes the roots of $x^p - t$. But since there is only one root, the only automorphism is the identity on L sending $t^{1/p} \mapsto t^{1/p}$.

9. (January 2016 Problem 5) Consider the group $\text{GL}_n(\mathbb{Q})$ of invertible $n \times n$ matrices with rational coefficients. Suppose $G \subset \text{GL}_n(\mathbb{Q})$ is a finite subgroup. Prove that every prime factor p of the order $|G|$ satisfies $p \leq n + 1$.

9. Solution Suppose that $p \mid |G|$, then by Cauchy's theorem there is an element $A \in G$ of order p , $A^p = I$. Notice that the minimal polynomial of A divides $x^p - 1$, so the minimal polynomial of A is either $x^p - 1$ or $\Phi_p(x)$. Since the minimal polynomial divides the characteristic polynomial, the degree of the minimal polynomial is at most n . Thus, if the minimal polynomial is $x^p - 1$ then we get $p \leq n$, and if the minimal polynomial is $\Phi_p(x)$ we get $p - 1 \leq n$. Thus, in all cases $p \leq n + 1$.